Benyamin Tabarsi

PhD Candidate at NC State University / Computer Scientist

Website: BenyaminTabarsi.com
Email: btaghiz@ncsu.edu
Phone: (984) 330-0340
Scholar: tinyurl.com/BTTscholar

Research and Teaching Profile

My work sits at the intersection of artificial intelligence, computing education, and human–computer interaction. I design and evaluate AI-powered tools, examine how students, educators, and software developers use and shape these technologies, and investigate novice programmers' struggles, planning, and scaffolding needs. In research, I combine system building, user studies, educational data mining, mixed methods, and participatory design. My broader research agenda is to advance human–AI co-intelligence in which human expertise guides what AI contributes and AI helps humans accomplish more and develop their own knowledge and skills. In teaching, I prioritize helping students build evaluative judgment and problem-solving skills through evaluation-first learning, collaborative reasoning, and scaffolding.

Highlights

Research and Publications

- 16 peer-reviewed publications (8 first-authored): 8 conference papers, 2 journal articles, 2 workshop papers, and 4 posters/demos; 6 manuscripts under review and 4 in preparation.
- Venues include ICER, VL/HCC, AAAI, SIGCSE, AIED, FIE, AI-HCI, CSEDM, Empirical Software Engineering, and Cluster Computing.
- Best Research Paper Award, IEEE VL/HCC 2024 (Jigsaw); Best Technical Demonstration Award, AAAI 2025 (MerryQuery).
- Designed and deployed MerryQuery, an LLM-powered learning assistant used by 750+ students across three semesters; lead researcher and coordinator for ExLAIM, an NSF ExLENT-funded program ($1M; NSF #2526340) pairing student developers with K–12 teachers.

Funding and Fellowships

- More than four years of continuous research assistantships on NSF CCRI, Creative CS, ExLENT, and RET projects.
- Contributed to the preparation of an $8,000 NC State DELTA grant to expand MerryQuery.
- Preparing the Professoriate Fellow (2025–2026) and Mentored Teaching Fellowship ($1,000), NC State University.

Teaching and Mentoring

- Instructor for Introduction to Computing Environments and a K–12 teacher workshop series on classroom LLM use; co-instructor for a 250+ student Introduction to Programming course; teaching assistant for three courses.
- Mentored 20 undergraduate researchers (from nine institutions across five states), 3 master's students, and 11 K–12 teachers; 14 mentees have co-authored published or submitted work.

Service and Outreach

- Program Committees: SIGCSE, ICER, ITiCSE, AAAI, IJCAI, KDD, LAK, AIED, FIE, and EDUCON.
- Journal Reviewer: Empirical Software Engineering and IEEE Transactions on Learning Technologies.
- Talk Series Coordinator, SPLICE LLMs Working Group (2024–present)
- 10 talks, panels, workshops, and presentations, plus a podcast appearance, an interview, and a news feature, primarily on AI in education and research training.

Education

North Carolina State University
Raleigh, NC

Ph.D. in Computer Science
Expected May 2027
Dissertation: Toward Human-AI Co-Intelligence in Computing Education: Evidence, Tool Design, and Interventions
Committee: Dr. Tiffany Barnes (Advisor), Dr. Dongkuan (DK) Xu (Co-Advisor), Dr. Lina Battestilli, Dr. Joey Huang

Science and Research Branch Azad University
Tehran, Iran

M.S. in Computer Software Engineering
2019

University of Mazandaran
Babolsar, Iran

B.S. in Information Technology Engineering
2016

Research Experience

Graduate Research Assistant
Game2Learn and GIC Labs
North Carolina State University
May 2022 – Present

- Lead MerryQuery, a trustworthy LLM-powered assistant that provides personalized support for students and instructors; deployed across three semesters of an introductory programming course (750+ students), supported by an $8,000 DELTA grant, and recognized with the AAAI 2025 Best Technical Demonstration Award.
- Lead ExLAIM, an NSF ExLENT-funded program ($1M; NSF #2526340) in which student developers and K–12 teachers co-design classroom AI tools; co-developed and co-taught a three-iteration professional development series on ChatGPT for K–12 teachers (FIE 2024, AI-HCI 2026).
- Study how people learn and work with generative AI: early-adopter developers' LLM use (Empirical Software Engineering, 2027), students' help-seeking with GenAI chatbots (AIED 2026), a systematic review of student-facing LLM tools (under review), HelpLens, a framework and dataset comparing teaching-staff and LLM responses to programming help requests, and SafeTalk, an AI coaching system for parent–teen health conversations.
- Modeled novice programmers' struggle and scaffolding needs from trace-log and think-aloud data (ICER 2024, VL/HCC 2023, CSEDM 2022) and contributed to novice programming tools including Jigsaw (VL/HCC 2024 Best Research Paper) and Pinpoint (VL/HCC 2022).

Research Assistant
Distributed Systems Lab
Azad University
Nov 2017 – Aug 2019

- Studied latency–power trade-offs and resource allocation in fog-based IoT networks.
- Guided students in conducting research and led group discussions on topics concerning distributed systems, IoT, and e-healthcare.

Peer-Reviewed Publications[1]

Conference Papers

[8] Heidi Reichert, Daniel Briceno*, **Benyamin Tabarsi**, Tiffany Barnes (2026). "Human-Centered Design of LLM-Powered Educational Chatbots: A Study with Secondary Teachers." In *Artificial Intelligence in HCI: 7th International Conference, AI-HCI 2026 (LNCS 16746), pp. 516–535.* [Link]

[7] Sam Gilson, **Benyamin Tabarsi**, Tiffany Barnes (2026). "Adversarial Assignment Perturbation: Effects on Help-Seeking Behaviors in Student Generative AI Chatbot Use." In *Proceedings of the 27th International*

[1] Students and teachers I mentored are indicated by an asterisk (*).

Conference on Artificial Intelligence in Education (AIED 2026), Part V, pp. 440–449. Overall acceptance rate: 24.8% (308/1,241). [Link]

[6] Heidi Reichert, **Benyamin Tabarsi,** Zifan Zang, Cheri Fennell*, Indira Bhandari, David Robinson, Madeline Drayton, Catherine Crofton, Matthew Lococo*, Dongkuan (DK) Xu, Tiffany Barnes (2024). "Empowering Secondary School Teachers: Creating, Executing, and Evaluating a Transformative Professional Development Course on ChatGPT." In *Proceedings of the 2024 IEEE Frontiers in Education (FIE 2024).* [Link]

[5] Heidi Reichert, **Benyamin Tabarsi,** Thomas Price, Tiffany Barnes (2024). "Jigsaw: A Tool for Decomposing and Planning Programming Problems." In *Proceedings of the 2024 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC). Overall acceptance rate: 32%.* ***Best Research Paper Award.*** [Link]

[4] **Benyamin Tabarsi,** Heidi Reichert, Nicholas Lytle, Veronica Cateté, Tiffany Barnes (2024). "Scaffolding Novices: Analyzing When and How Parsons Problems Impact Novice Programming in an Integrated Science Assignment." In *Proceedings of the 2024 ACM Conference on International Computing Education Research (ICER).* Acceptance rate: 20.5% (36/176). [Link]

[3] Wengran Wang, John Bacher, Amy Isvik, Ally Limke, Sandeep Sthapit, Yang Shi, **Benyamin Tabarsi**, Keith Tran, Veronica Cateté, Tiffany Barnes, Chris Martens, Thomas Price (2023). "Investigating the Impact of On-Demand Code Examples on Novices' Open-Ended Programming Experience." In *Proceedings of the 2023 ACM Conference on International Computing Education Research (ICER).* Acceptance rate: 21.3% (35/164). [Link]

[2] **Benyamin Tabarsi,** Heidi Reichert, Rachel Qualls*, Thomas Price, Tiffany Barnes (2023). "Exploring Novices' Struggle and Progress During Programming Through Data-Driven Detectors and Think-Aloud Protocols." In *Proceedings of the 2023 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC).* [Link]

[1] Wengran Wang, Gordon Fraser, Mahesh Bobbadi, **Benyamin Tabarsi,** Tiffany Barnes, Chris Martens, Shuyin Jiao, Thomas Price (2022). "Pinpoint: A record, replay, and extract system to support code comprehension and reuse." In *Proceedings of the 2022 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC).* Acceptance rate: 29.3% (12/41). [Link]

Journal Articles

[2] **Benyamin Tabarsi**[†], Heidi Reichert[†], Sam Gilson, Ally Limke, Sandeep Kuttal, Tiffany Barnes (2027). "LLMs' reshaping of people, processes, products, and society in software development: a qualitative exploration with early adopters." *Empirical Software Engineering, 32(1), ([†]Equal contribution) Article 21.* SJR Q1 (2025). [Link]

[1] **Benyamin Tabarsi**, Ali Rezaee, Ali Movaghar (2023). "ROGI: Partial Computation Offloading and Resource Allocation in the Fog-Based IoT Network Towards Optimizing Latency and Power Consumption." *Cluster Computing, 26(3), 1767–1784.* SJR Q1 (2025). [Link]

Workshop Papers

[2] **Benyamin Tabarsi**, Ally Limke, Heidi Reichert, Rachel Qualls*, Thomas Price, Chris Martens, Tiffany Barnes (2022). "How to Catch Novice Programmers' Struggles: Detecting Moments of Struggle in Open-Ended Block-Based Programming Projects using Trace Log Data." In *Proceedings of the 6th Educational Data Mining in Computer Science Education (CSEDM) Workshop.* [Link]

[1] Heidi Reichert, Ally Limke, **Benyamin Tabarsi**, Thomas Price, Chris Martens, Tiffany Barnes (2022). "How, when, and why do novices struggle in programming? Exploring the experiences and perceptions of common programming moments in block-based environments." In *Proceedings of the 6th Educational Data Mining in Computer Science Education (CSEDM) Workshop.* [Link]

Posters and Demonstrations

[4] **Benyamin Tabarsi** (2026). "MerryQuery in the Loop: How Students Use and Perceive a GenAI-Powered Assistant in an Introductory Programming Course." In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education, Volume 2 (SIGCSE).* [Link]

[3] **Benyamin Tabarsi** (2025). "Developing LLM-Powered Trustworthy Agents for Personalized Learning Support." In

Proceedings of the AAAI Conference on Artificial Intelligence, 39(28), Doctoral Consortium. [Link]

[2] **Benyamin Tabarsi**, Aditya Basarkar*, Xukun Liu, Dongkuan (DK) Xu, Tiffany Barnes (2025). "MerryQuery: A Trustworthy LLM-Powered Tool Providing Personalized Support for Educators and Students." In *Proceedings of the AAAI Conference on Artificial Intelligence, 39(28).* ***Best Technical Demonstration Award.*** [Link]

[1] Heidi Reichert, Sandeep Sthapit, **Benyamin Tabarsi,** Ally Limke, Thomas Price, Tiffany Barnes (2024). "Experience Helps, but It Isn't Everything: Exploring Causes of Affective State in Novice Programmers." In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education, Volume 2 (SIGCSE).* [Link]

Papers in Progress

Under Review

[6] **Benyamin Tabarsi**, Michelle Jiang*, Xiaoyi Tian, Dongkuan (DK) Xu, Lina Battestilli, Tiffany Barnes. "One Prompt Away: A Semester of High-Assistance, Course-Grounded GenAI Help in Introductory Programming for Engineering Students." Under review.

[5] Joey Huang, Jiwon Lee, **Benyamin Tabarsi**, Xiaoyi Tian, Dongkuan (DK) Xu, Tiffany Barnes. "Different Starting Points, Different Roles: Teacher Learning in Co-Designing Generative AI Tools." Under review.

[4] Wenbo Li, **Benyamin Tabarsi**, Michelle Jiang*, Dongkuan (DK) Xu, Tiffany Barnes. "AI Coaching for Parent-Teen Sexual Health Communication: From Barriers to Design Implications." Under review.

[3] Michelle Jiang*, **Benyamin Tabarsi,** Tiffany Barnes. "Domain Shift in Knowledge Tracing for Students in Opportunity Zones." Under review.

[2] Thomas Price, Samiha Marwan, Inaas Asad, **Benyamin Tabarsi**, Keith Tran, Tommy Bacher, Tiffany Barnes, Shuyin Jiao, Michael Liut, Andrew Petersen. "Evaluation of Adaptive Immediate Feedback for Programming: A Multi-Study Conceptual Replication." Under review.

[1] **Benyamin Tabarsi**, Tahreem Yasir, Heidi Reichert, Xiaoyi Tian, Shiva Gadireddy*, Clara DiMarco, Daniel Briceno*, Lina Battestilli, Tiffany Barnes. "Student-Facing LLM Tools in Computing Education: A Systematic Literature Review of Roles, Designs, Evaluations, and Impacts." Under review [Link].

In Preparation

[4] **Benyamin Tabarsi**, Deniz Ozturk, Carrie Robledo, Joanna Perkins, Xiaoyi Tian, Shiyan Jiang, Tiffany Barnes. "Elementary Students' Conceptions and Designs of Compassionate AI in a Teacher-Led Classroom Challenge." *To be submitted.*

[3] **Benyamin Tabarsi**, Sam Gilson, Faaz Mohamed*, Matheus Marques*, Lokesh Sharma*, Susanna Quayle*, Nicole Ackerman-Lowery*, Daniel Briceno*, Aditya Basarkar*, Xiaoyi Tian, Joey Huang, Dongkuan (DK) Xu, Tiffany Barnes. "Teachers and Developers Building Together: The ExLAIM Six-Week Co-Design Program for Developing AI-Supported Classroom Tools." *To be submitted.* [Link]

[2] Wenbo Li, **Benyamin Tabarsi**, Tahreem Yasir, Aryan Santhosh Kumar*, Laura Widman, Dongkuan (DK) Xu, Tiffany Barnes. "SafeTalk: Designing and Evaluating an AI-Based Coaching Platform for Parent–Teen Sexual Health Conversations." *To be submitted.* Earlier version: arXiv:2602.00017.

[1] **Benyamin Tabarsi**, Michelle Jiang*, Susanna Quayle*, Xiaoyi Tian, Jayani Sivakumar*, Lina Battestilli, Thomas Price, Dongkuan (DK) Xu, Tiffany Barnes. "Characterizing Teaching Staff and LLM Responses to Forum Queries in Introductory Programming: A Dataset and Pedagogical Framework." *To be submitted.*

Selected Research Systems and Artifacts

▪ **MerryQuery: GenAI-Powered Educational Assistant (Link)**
Led the design and evaluation of an LLM-powered assistant deployed across three semesters of introductory MATLAB (750+ students). The deployment produced evidence on how students use and perceive GenAI help in an

introductory course (SIGCSE 2026), and the system earned the AAAI 2025 Best Technical Demonstration Award.

▪ **HelpLens: Pedagogical Response-Analysis Framework and Dataset**
Developed a theory-informed, eight-dimension framework for analyzing teaching-staff and LLM responses to programming help requests. Applied it to 172 paired responses to create an annotated dataset and characterize pedagogical response patterns.

▪ **SafeTalk: AI Coaching System for Parent–Teen Sexual Health Communication (Link)**
Mentored the design and evaluation of an expert-informed AI coaching system that lets parents rehearse difficult conversations with their teens before having them.

Honors and Awards

▪ **Preparing the Professoriate (PTP) Fellow**, *NC State University* 2025 – 2026
Selected for a competitive, yearlong program in pedagogy, faculty mentorship, and mentored co-teaching under Dr. Lina Battestilli in CSC 113 (Introduction to Programming in MATLAB).

▪ **Best Technical Demonstration Award at AAAI 2025**
For "MerryQuery: A Trustworthy LLM-Powered Tool Providing Personalized Support for Educators and Students"

▪ **Best Research Paper Award at IEEE VL/HCC 2024**
For "Jigsaw: A Tool for Decomposing and Planning Programming Problems"

▪ **Ranked First in Master's Cohort by Cumulative GPA**
Highest cumulative GPA among students in the entering 2016 master's cohort in Computer Software Engineering.

Funding and Fellowships

▪ **NSF-Funded Research Assistantships,** *NC State University* 2022 – Present
Supported through more than four years of continuous research assistantships on NSF-funded Game2Learn projects, including NSF CCRI (#2213792), NSF Creative CS (#1917885), ExLENT (#2526340), and RET (#2055528); PIs: Thomas Price and Tiffany Barnes.

▪ **Mentored Teaching Fellowship ($1,000)**, *NC State University* Spring 2026
Awarded for co-instruction under Dr. Lina Battestilli's mentorship.

▪ **SIGCSE Travel Grant ($360)**, *NC State University* Spring 2026
Travel grant for participation in SIGCSE 2026.

▪ **DELTA Grant**, *NC State University* 2025 – 2026
Contributed to the preparation of an $8,000 DELTA grant, awarded with Dr. Tiffany Barnes, Dr. DK Xu, and Dr. Lina Battestilli, to expand MerryQuery development.

▪ **AAAI 2025 Student Travel Grant ($500)**
Travel grant for participation in the AAAI 2025 doctoral consortium and demonstration tracks.

▪ **Computer Science Department Travel Fund ($460),** *NC State University*
Supplementary travel funding for participation in AAAI 2025.

▪ **SPLICE LearnLab Summer School Grant (2023–2024)**
Supported participation in the Educational Data Mining (2023) and Computing Education Research (2024) tracks at Carnegie Mellon University's LearnLab Summer School.

Teaching Experience

▪ **Discrete Mathematics, CSC 226** (Teaching Assistant), *NC State University* May – Aug 2026
Delivered three lectures and supported 22 students through grading and office hours.

▪ **Introduction to Programming** (Co-Instructor), *NC State University* Jan – May 2026
Delivered selected lectures, co-led weekly TA meetings, and designed active-learning activities and responsible

GenAI-use guidance in a 250+ student MATLAB course.

▪ **How to Integrate LLMs into K–12 Classrooms** (Instructor and Mentor)**,** *NC State University* Jun 2023 – Jun 2024
Mentored the development and co-led instruction of a workshop series for K–12 teachers on integrating ChatGPT into classroom instruction across three iterations.

▪ **Introduction to Computing Environments** (Instructor)**,** *NC State University* May – Aug 2023
Administered E 115 for 58 students; coordinated curriculum implementation and student support; and supervised teaching assistants.

▪ **Art, Coding, Action!** (Teacher and Curriculum Consultant)**,** *NC State University* Jul 2022
Led 20+ high school students in a one-week block-based programming camp, focusing on game and art projects in Snap!

▪ **Automated Learning and Data Analysis** (Teaching Assistant)**,** *NC State University* Jan 2022 – May 2022
Designed coding and written questions, graded coursework, and held office hours for approximately 100 students in CSC 422/522.

▪ **Data Structures and Algorithms** (Teaching Assistant)**,** *NC State University* Aug 2021 – Dec 2021
Graded assignments and exams and provided academic support during office hours.

Talks, Panels, and Workshops

▪ **ExLAIM** - Poster, NC State Applied AI in Engineering & Computer Science Symposium Fall 2026
Presented ExLAIM's six-week co-design program, in which K–12 teachers and student developers build AI-supported classroom tools together.

▪ **Toward Human-AI Co-Intelligence in Computing Education** - Guest Talk Spring 2026
Presented evidence synthesis, industry perspectives, tool design, interventions, and human–AI response evaluation to support human–AI collaboration in NC State's "Advanced Topics in Efficient Deep Learning" course.

▪ **MerryQuery** - Guest Talk, Utah State University Fall 2025
Discussed AI in education and the status of MerryQuery in a graduate Natural Language Processing course.

▪ **MerryQuery** - Guest Talk, NC State DELTA AI Roundtable Fall 2025
Presented early MerryQuery findings in the session "IDIG: AI in Education."

▪ **Graduate Life and PhD Experience** - Panel Fall 2025
Discussed graduate school and the future of computing in the age of AI.

▪ **MerryQuery** - Poster, NC State Applied AI in Engineering & Computer Science Symposium Fall 2024
Presented MerryQuery's design and early classroom deployment with Dongkuan (DK) Xu and Tiffany Barnes.

▪ **Developing Effective Research Questions** - Workshop Summer 2023 and 2024
Taught undergraduate research interns to develop focused, feasible research questions.

▪ **PhD in CS Experience** - Panel Spring 2024
Discussed the PhD experience and strategies for success at NC State Doctoral Recruiting Day.

▪ **How to Present Research** - Workshop Summer 2023
Guided undergraduate research interns on effective poster and oral presentation techniques.

▪ **From the Internet of Things to Fog Computing** - Guest Lecture 2018
Presented the evolution of distributed systems from IoT to fog computing for students in a Distributed Systems course.

Media Coverage

▪ **AI in Education** - Podcast Summer 2025
Discussed opportunities and challenges of AI in education and MerryQuery's development and research journey.

▪ **Computing Education and Generative AI** - AIhub Interview Aug 2025
Interviewed about MerryQuery and research at the intersection of generative AI, computing education, and human-centered AI.

▪ **AAAI 2025 Demonstration Award** - News Feature Spring 2025
Featured by the NC State Department of Computer Science following the AAAI-25 Best Technical Demonstration Award.

Mentorship Experience

Outcomes
14 mentees have co-authored published or submitted work; mentees have come from nine universities and community colleges across five states. Two mentees began Ph.D. and one community-college mentee transferred to NC State and continued on the project.

Undergraduate Research Interns
MerryQuery and ExLAIM
1. Faaz Mohamed (*University of Pittsburgh, PA)* *Summer 2026*
2. Matheus Marques (*The College of New Jersey, Ewing, NJ)* *Summer 2026*
3. Lokesh Sharma (*University of California, Irvine)* *Summer 2026*
4. Susanna Quayle (*NC State University, Raleigh, NC)* *Summer 2026*
5. Nicole Ackerman-Lowery (*Fayetteville Technical Community College, Fayetteville, NC)* *Summer 2026*
6. Daniel Briceno (NC State University, Raleigh, NC) Spring 2026, Summer 2026
7. Aditya Basarkar (NC State University, Raleigh, NC) Summer 2024, Summer 2026
8. Aishwarya Radhakrishnan (NC State University, Raleigh, NC) *Summer 2024, Fall 2024*
9. Praneel Magapu (NC State University, Raleigh, NC) *Summer 2024*
10. Lavan Aditya (NC State University, Raleigh, NC) *Summer 2024*

HelpLens
1. Jayani Sivakumar (*NC State University, Raleigh, NC)* *Summer 2025*
2. Susanna Quayle (*Fayetteville Technical Community College, Fayetteville, NC)* *Spring 2025, Summer 2025*

SafeTalk
1. Aryan Santhosh Kumar (*NC State University, Raleigh, NC*) Fall 2024, Spring 2025
2. Michelle Jiang (*University of North Carolina at Chapel Hill, NC*) Summer 2024
3. Jonathan Hardwick (*Fayetteville Technical Community College, Fayetteville, NC*) Summer 2024
4. Susanna Quayle (*Fayetteville Technical Community College, Fayetteville, NC*) Summer 2024

Systematic Literature Review on LLMs and Computing Education
1. Shiva Gadireddy (*NC State University, Raleigh, NC*) Summer 2023, Fall 2023, Spring 2024
2. Lavan Aditya (*NC State University, Raleigh, NC*) Spring 2024
3. Samantha Gonzalez (*Kean University, Union, NJ*) Summer 2023
4. Yadhira Marcos-Avila (University of North Carolina at Charlotte, NC) Summer 2023

Detectors for Identifying Novices' Struggle during Programming
1. Rachel Qualls (University of Alabama, Tuscaloosa, AL) Summer 2022

Participatory Design on a Block-Based Programming Learning Management System (SnapClass)
1. Maggie Lin (*NC State University, Raleigh, NC*) Summer 2022
2. Sana Mahmoud (*NC State University, Raleigh, NC*) Summer 2022

Master's Students
MerryQuery
1. Shreyas Visweshwaran (*NC State University, Raleigh, NC*) Fall 2025
2. Teddy Chen (*NC State University, Raleigh, NC*) Fall 2024
3. Homak Patel (*NC State University, Raleigh, NC*) Fall 2024

Teacher Research Interns

MerryQuery and ExLAIM

1. Algie Brodnax (*GTCC Middle College, NC*) Summer 2026
2. Latoya Clay (*Hillside High School, NC*) Summer 2026
3. Lohanna Sanchez (*Ragsdale High School, NC*) Summer 2026
4. Madeline Cutchin (*Nash Central Middle School, NC*) Summer 2026
5. Marcia Winborne (*JF Webb High School, NC*) Summer 2026
6. Marie Antonette Jimenez (*Discovery Charter School, NC*) Summer 2026
7. Mason McDaniel (*Broughton High School, NC*) Summer 2026
8. Tiffany McMillan (*Jack Britt High School, NC*) Summer 2026
9. Tina Fox (*Apex Friendship High School, NC*) Summer 2026

Integrating LLMs into K–12 Classrooms

1. Cheri Fennell (*Durham Public Schools, NC*) Summer 2023
2. Matthew Lococo (*Greene County School, NC*) Summer 2023

Professional Service and Memberships

Program Committee Member

ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE)	2026
ACM Conference on International Computing Education Research (ICER)	2026
AAAI Conference on Artificial Intelligence	2026
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)	2025 - 2026
International Joint Conference on Artificial Intelligence (IJCAI)	2025
IEEE Frontiers in Education Conference (FIE)	2024 - 2025
International Conference on Artificial Intelligence in Education (AIED)	2025
AAAI 2025 Spring Symposium: GenAI@Edge: Empowering Generative AI at the Edge	2025
ACM International Conference on Learning Analytics & Knowledge (LAK)	2025
IEEE Global Engineering Education Conference (EDUCON)	2025
ACM Technical Symposium on Computer Science Education (SIGCSE)	2023 - 2025

Journal Reviewer

Empirical Software Engineering
IEEE Transactions on Learning Technologies

Community Leadership

- **Talk Series Coordinator**, SPLICE Large Language Models Working Group Sep 2024 - Present

Organize the working group's monthly virtual talk series on LLMs in computing education: recruit and invite speakers, coordinate logistics with presenters, and moderate live sessions. Hosted 13 sessions featuring 19 presenters.

Memberships

- AAAI (2025–present); IEEE (2018; 2023–present); ACM (2023–present); Sigma Xi (2025–2026)

Skills

Programming: Python, Java, C++, R, MATLAB, HTML/CSS, PHP
Tools & Libraries: PyTorch, scikit-learn, LangChain, LlamaIndex, Git, Docker
Database & Operating Systems: MySQL, MongoDB, Linux, PostgreSQL
Research Methods: mixed-methods studies, experimental and quasi-experimental design, qualitative interviewing and coding, surveys, human-centered and participatory design, educational data mining, systematic literature reviews, nonparametric statistics
GenAI: retrieval-augmented generation (RAG), multi-agent systems, LLM guardrails, LLM evaluation (LLM-as-a-judge, human evaluation protocols, NLI-based consistency checks)